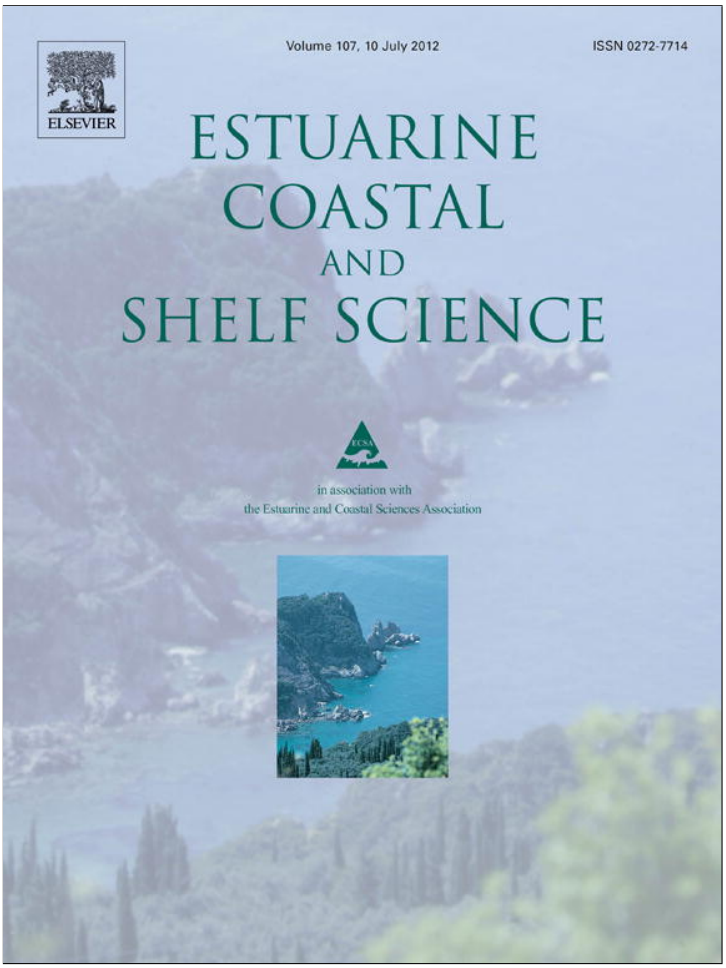
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Predicting fish community properties within estuaries: Influence of habitat type and other environmental features

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ABSTRACT

Statistical models predicting species distributions are essential not only to increase knowledge on species but for their application in conservation and ecologically-based management. The variation of fish species richness and abundance in the most representative habitats (saltmarsh, mudflat and subtidal) in five estuaries along the Portuguese coast was analysed through seasonal sampling surveys in 2009. Generalized additive models (GAM) were developed to describe the variation of species richness and abundances with a set of geomorphologic, hydrologic and environmental characteristics from the sampled estuaries and habitats. GAM were chosen as the complex interactions dominating these ecosystems and species distribution are non-linear. Final models built for each estuary and for all estuaries together performed well during the calibration phase and also during the validation phase, where an unused data sub-set from each estuary was used. There was not a similar combination of variables retained by the models for the studied estuaries but factors such as the area of the habitat, the distance to estuary mouth, percentage of mud in the sediment and depth were commonly retained. The partial effect of these predictor variables on the variation of species richness and abundance in the estuaries varied markedly and the importance of preserving the heterogeneity of habitats within estuaries was highlighted. Models for each individual estuary performed better than models for estuaries combined. Predictive models could be useful as a preliminary tool to prepare long-term conservation plans at different scales.

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1. Introduction

Estuaries are one of the most productive ecosystems on Earth (Costanza et al., 1997), and have been the subject of considerable ichthyological research, some of which specifically focussing on the key environmental factors affecting the structure of estuarine fish communities (Selleslagh et al., 2009). These systems are exposed to great environmental variability, with the life cycle of most of organisms showing clear seasonal patterns in growth, reproduction and abundance (Coma et al., 2000; Maes et al., 2004). Understanding the relationships between environmental factors and fish species distribution in estuaries is of fundamental importance, and attaining quantitative predictions of the patterns of occurrence and abundance of fish species in estuaries represents an essential goal for its application in reserve planning (Corsi et al., 1999; Austin, 2002; Elith et al., 2006) or biodiversity conservation (Gray et al., 2006; Lira-Noriega et al., 2007; Ko et al., 2008).

The quantification of such species environment relationships represents the core of predictive geographic modelling in ecology (Guisan and Zimmermann, 2000). Many predictive models based on correlating presence and abundance data of species with environmental predictors have been widely used (Manel et al., 1999; Ko et al., 2008) through a variety of statistical methods, often in conjunction with Geographic Information Systems (GIS) and remote-sensing (Austin, 2002).

Within coastal and marine ecosystems recent attempts have been made to statistically model the responses of fish species to environmental variables on a large scale and to use these models to predict their distribution and occurrence (Nicolas et al., 2010; França et al., 2011). Nevertheless, there are increasing demands in order to obtain reliable and quantitative predictive tools, as they are required to interpret species preferences and tolerances to current environmental conditions, as well as species response to anticipated changes in environmental conditions (Ysebaert et al., 2002).

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Research on predictive techniques applicable to species distribution is still far from producing a reliable modelling system (Guisan and Zimmermann, 2000; Hirzel and Guisan, 2002).

Traditionally, models used in ecology to predict potential species distributions were multivariate in nature and based on linear functions, however, they led to some statistical and theoretical concerns, and methods able to model non-linear relationships are now available (Ko et al., 2008). These models are better tools towards accurately predicting species distribution (Stockwell, 2007). As the number and complexity of these models evolve, difficulty over which one to choose represents an increasing challenge (Ko et al., 2008). Recently, several reviews, methodological comparisons and interpretations on the value, use and application of the different methods have been made (Austin, 2007). These pointed out that uncertainty still exists as to which model is the best to select under different spatial scales and/or species characteristics (Zaniewski et al., 2002).

Recent loss of estuarine habitat due to constant demands from a wide range of human activities in coastal areas has resulted in an increased attention by managers and researchers towards the importance of fish habitats. Concerns about severe habitat loss and degradation in estuaries have prompted managers to identify, prioritize and protect essential habitats for estuarine organisms (Bacheler et al., 2009; França et al., 2012). Habitat distribution models are among the available tools that allow the linking of species distribution to environmental conditions (Zuchetta et al., 2010). Such models use quantitative methods to infer species environmental requirements from conditions at known occurrences (Guisan and Zimmermann, 2000). Some authors have also developed habitat distribution models aimed at identifying fish nursery habitats; for example for flatfish, both in coastal waters and in estuaries or lagoons (Stoner et al., 2001; Manderson et al., 2002; Eastwood et al., 2003; Le Pape et al., 2003, 2007; Nicolas et al., 2007). As starting point, a basic understanding of habitat use by the species of interest is required for all of these techniques (Bacheler et al., 2009).

Estuaries along the Portuguese coast have been intensively studied in recent years, and different dimensions of their nursery role and habitat use for juveniles of several commercially important fish species evaluated (e.g. Cabral et al., 2007; Leitão et al., 2007; Pombo et al., 2007; Vasconcelos et al., 2010). Furthermore, the spatial patterns of habitat use by fish assemblages in these estuaries have shown that in addition to saltmarsh habitats, that generally supported elevated densities of fish, particular species were found to be associated with specific habitat types, highlighting the important role different habitats play in the functioning of these estuarine ecosystems (França et al., 2009).

The aim of the present study was to develop statistical models (generalized additive models) to predict variation in fish species richness as well as abundance in different estuarine habitats based on environmental features. The variation of these environmental variables within each system, and at different scales, as well as the robustness and reliability of the models were also evaluated.

2. Material and methods

2.1. Study area

Five estuarine systems along the Portuguese coast were considered in this study: Ria Aveiro, Tejo, Sado, Mira and Guadiana (Fig. 1). These systems differ considerably in terms of their geomorphologic and hydrologic characteristics (Table 1): Tejo and Sado present the largest areas with 320 km^2 and 180 km^2 respectively, and Mira is the smallest with 5 km^2. The Tejo estuary has considerably higher mean river flow values (300 $m^3 s^{-1}$) as well as larger estuary mouth width (5.3 km) than the other estuaries. In addition, anthropogenic pressure index according to Vasconcelos et al. (2007) also varies, ranging from 0.14 to Mira estuary to 0.76 for the Tejo estuary, the most pressured system. Moreover, habitat complexity (score based on the structure and patchiness of the habitats present in the estuary and their areas, with higher scores attributed to estuaries with more complex habitat structures and larger areas of the different habitats) can be considered high in Ria de Aveiro (score 3), medium in Tejo and Sado (2) and low in Mira and Guadiana (score 1). Shallow areas predominate in all the estuarine systems as mean depths vary between 1 and 6 m (Table 1).

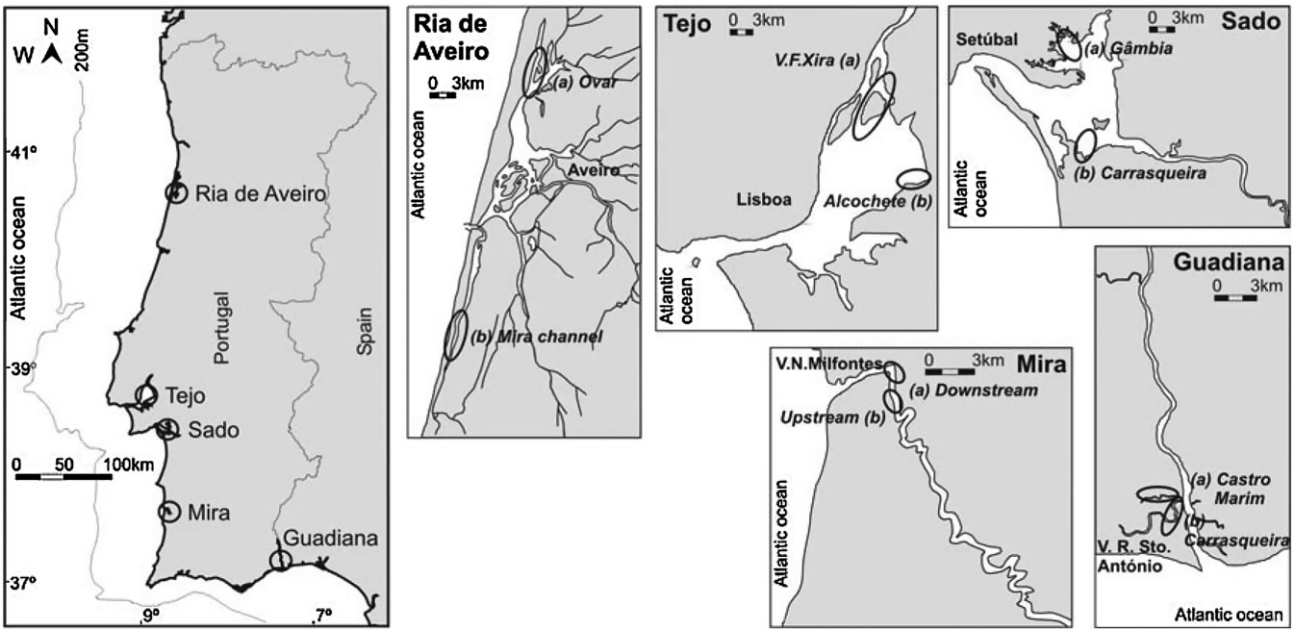


Fig. 1. Estuarine systems sampled in the Portuguese coast. Also shown is the location of sites within each estuary where the three habitats (saltmarsh, mudflat and subtidal) were sampled.

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Table 1
Main geomorphologic and hydrologic characteristics of estuaries: area, river flow, estuary mouth width, anthropogenic pressure index [determined in Vasconcelos et al. (2007) and ranging from 0 – lowest to 1 – highest pressure)] and estuary's habitat complexity (1 – low complexity, 2 – medium complexity and 3 – high complexity). Also shown, features of habitats sampled (habitat area and distance to estuary mouth) and annual mean (and standard deviation) of environmental variables measured in seasonal sampling surveys in 2009 in those habitats (temperature, salinity, depth, organic matter content in the sediment and mud content in the sediment).

Estuary	Area (km^2)	Riverflow ($m^3\ s^{-1}$)	Mouth width (km)	Pressure index	Complexity	Habitat	Habitat area (km^2)	Distance to mouth (km)	Temp (°C)	Salinity	Depth (m)	Organic matter	Mud (%)
Ria Aveiro	74	40	0.43	0.42	3	Mudflat	2.5(0.3)	18.5(4.3)	17.9(5.4)	18.8(9.7)	0.9(0.3)	1.5(1.6)	17.9(23.2)
						Saltmarsh	8.1(0.3)	18.5(4.3)	18.1(6.2)	20.3(9.4)	1.1(0.5)	7.9(8.3)	33.9(24.2)
						Subtidal	5.0(0.3)	17.3(4.5)	16.9(5.0)	21.3(7.9)	2.7(1.4)	3.5(3.6)	30.2(36.7)
Tejo	320	300	5.3	0.76	2	Mudflat	14.4(10.9)	35.4(1.9)	17.3(4.9)	17.4(7.6)	1.8(1.3)	9.4(1.3)	97.6(4.3)
						Saltmarsh	2.9(0.6)	36.6(2.4)	16.0(5.0)	17.8(7.2)	1.2(0.9)	9.9(5.5)	96.6(10.2)
						Subtidal	32.3(7.7)	35.9(1.8)	17.0(5.5)	18.0(8.4)	3.4(1.4)	6.6(3.8)	63.5(33.2)
Sado	180	40	1.91	0.49	2	Mudflat	16.7(9.3)	19.7(2.2)	20.4(5.4)	34.8(5.6)	1.5(0.3)	9.5(8.3)	59.7(30.4)
						Saltmarsh	3.3(1.5)	20.3(2.2)	17.5(7.3)	34.8(1.0)	1.0(0.5)	10.1(5.1)	61.5(20.7)
						Subtidal	5.0(4.0)	19.1(2.8)	18.5(6.3)	34.5(3.8)	4.2(1.8)	7.2(3.7)	79.9(28.2)
Mira	5	3	0.57	0.14	1	Mudflat	3.1(0.3)	4.7(1.0)	16.6(2.8)	33.2(4.8)	1.9(1.5)	10.6(2.3)	83.4(11.0)
						Saltmarsh	0.6(0.1)	4.8(1.0)	16.7(2.7)	23.7(4.6)	1.3(0.4)	11.8(2.8)	90.6(4.9)
						Subtidal	0.3(0.1)	4.9(1.0)	16.9(2.5)	35.0(1.7)	5.4(3.0)	8.4(1.4)	64.0(26.6)
Guadiana	20	80	1.76	0.21	1	Mudflat	0.6(0.1)	6.3(0.4)	17.2(3.8)	31.7(5.8)	1.6(0.6)	6.4(0.9)	69.6(13.1)
						Saltmarsh	5.6(0.1)	6.0(0.2)	17.9(3.4)	34.3(1.5)	1.5(0.3)	9.8(3.3)	80.9(12.5)
						Subtidal	1.5(0.6)	6.3(0.3)	18.6(3.5)	33.2(3.5)	2.9(0.5)	8.9(8.3)	54.1(18.5)

In each system two sites with similar salinity and depth were selected and considered sampling replicates. In both sites three habitats were sampled: saltmarsh (intertidal, sediment-based, macrophyte dominated, saline-influenced habitats); mudflats (areas of unvegetated intertidal habitats, lying between the highest and lowest tides, and composed predominantly of sediments from fine silt to coarse sands) and subtidal channels (permanently subtidal unvegetated habitats, composed predominantly of sediments ranging from fine silts to coarse sands). Present habitat classification was adapted from Pihl et al. (2002). Previously, the different habitat types of each estuarine system were mapped in order to assess habitat composition and variability (França et al., 2009). Habitat mapping was performed with a combination of aerial photographs, field observations and data from previous studies, using ArcGis 9 (ESRI Inc.). Sites and habitats sampled within each estuary in the present study were chosen based on the same work performed by França et al. (2009) to establish sampling replicates constituted by similar mosaics of the three habitats. ArcGis 9 (ESRI Inc.) was also used to determine the area of each habitat and their distance from to the estuary's mouth.

2.2. Sampling surveys

Fish assemblages were sampled seasonally in 2009 (winter, spring, summer and autumn). Sampling was done with a beam trawl in all habitat types. Three hauls were performed in each habitat, in both areas within each estuary, and trawls were restricted to each habitat type to avoid overlap between different habitats. A total of 72 hauls were performed for each estuary.

Sampling was conducted during the night, using a 2 m beam trawl with a tickler chain and 5 mm mesh size in the cod end. Trawls were towed at a constant speed (0.8 m s^{-1}) and lasted for 10 min whenever there was sufficient habitat area available. The trawl opening (2 m) and the distance travelled (obtained using the coordinates registered at the beginning and at the end of each trawl with a global positioning system (GPS)) allowed estimation of the area trawled. Although the water of most Portuguese estuaries is turbid, shallow areas (e.g. intertidal areas) might present high water transparency, and thus the night period allowed to reduce fish avoidance to the net by visual recognition (previous studies have analysed differences between day and night hauls and results taking into account for the selection of the best sampling period). At the beginning of each trawl, water salinity and temperature (°C) were measured with a multi-parameter probe (WTW) and depth (m) was also registered. In each sampled habitat three replicates of sediment were collected using a Van Veen grab (0.05 m^2) for the determination of mean mud content in the sediment (percentage of dry sediment not retained in a 0.063 mm calibrated sieve) and mean organic matter content in sediment (determined by weight loss on ignition of dry sediment during 4 h at 500 °C).

All fish caught were identified, counted and measured (total length with 1 mm precision).

2.3. Data analysis

2.3.1. Model calibration

Generalized additive models (GAM) (Hastie and Tibshirani, 1990) were used to build the models. They are semi-parametric extensions of Generalized Linear Models (GLM), are very flexible and allow both linear and complex additive response shapes, as well as combinations of the two within the same model (Wood and Augustin, 2002). GAM allows estimating response curves with a non-parametric smoothing function instead of parametric terms, which make them useful to model non-linear responses of data to environmental parameters (Wood and Augustin, 2002).

GAM were conducted using R software (R Development Core Team, 2005) using the "mgcv" library (Wood, 2000) for R software packages and some custom functions to implement the GAM. Species richness and species density were modelled separately but with an equivalent approach. Firstly and for each estuary, (1) models were designed in order to relate the variation in each estuary of fish species richness and densities in different estuarine habitats with several variables: sampling season (January, April, July and October) and habitat type (saltmarsh, mudflat and subtidal) were parameterised as classification variables in the model (in order to examine the effect of the different habitat and seasonal changes), area of each habitat type (km^2), distance from the sampled point to the estuary's mouth (km), water temperature (°C), salinity, depth (m) and percentage of organic matter and mud in the sediment. In addition, (2) models were also built considering data from all the estuaries, in order to relate the variation in fish species richness and densities in different habitats with their main geomorphologic and hydrologic characteristics. In this case, besides all the environmental variables used previously, the following were also used: total estuarine area (km^2), mean annual river flow ($m^3\ s^{-1}$), estuary mouth width, anthropogenic pressure index

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(according to Vasconcelos et al., 2007) and estuarine complexity value (based on the number of different habitat types present in the estuary and their areas).

The choice of the environmental variables was based on their ability to characterize ichthyofaunal assemblages in estuaries. The number of candidate variables included in the final model was firstly reduced by removing highly correlated variables. Following Fielding and Haworth (1995), a Spearman correlation analysis was conducted and variables with high correlation coefficient ($r > 0.8$) were not considered for model calibration, in order to avoid multicollinearity.

All possible combinations of the measured variables were analysed and the best fitted models were selected. Interactions between variables were not relevant and were therefore not included in the analysis. A Poisson distribution with a log link function was applied when modelling fish species richness and a Gamma distribution with a log link function for fish species densities. A cubic smoothing spline method was chosen to smooth the studied variables. The degrees of freedom for each individual smoothing component were specified and a maximum of 4 were used when fitting the models. Although this will reduce the amount of flexibility, a more conservative approach should be addressed when relatively small datasets are used.

2.3.2. Model selection and validation

In order to evaluate the quality of predictions for each community descriptor (fish species richness and densities), the original dataset was divided into two sub-sets: calibration and validation. The first sub-set, a random sample from 75% of the total database, was used to calibrate (train) the models, whereas the second, constituted by the remaining data (25%) was used to evaluate (test) the robustness of model predictions. Replicates of each "habitat × season" combination were represented in both datasets in order to make them coherent (the choice of the replicates for each combination was done randomly).

Generally, models are selected according to the Akaike Information Criterion (AIC; Burnham and Anderson, 2002), with the best fitted models presenting the smallest AIC. Nonetheless, the AIC can over-estimate the number of parameters needed in the model (Kass and Raftery, 1995), thus, in this case several criteria were used to choose the best model, as follows.

A set of candidate independent GAMs for each estuary, which explained high percentages of deviance and had small AIC values were chosen. As over-parameterization can lead to high deviance explained for calibration dataset, but low deviance explained for the validation dataset, each of these models was run with the validation (test) dataset. From this sub-set of models, the final one was chosen based on the highest percentage of deviance explained.

3. Results

3.1. Physical and environmental variables

In addition to the differences in geomorphologic and hydrological features of the sampled estuaries (Table 1; described in Section 2.1 Study area), sampled habitats in each site also presented variations with habitat area and distance between sampling point and estuary mouth varying mostly in agreement with total estuarine area: generally larger in Tejo and Sado and smaller in Mira and Guadiana.

3.2. Habitat fish communities

A total of 48 fish species were caught in the saltmarsh, mudflat and subtidal habitats of the five sampled estuaries. The mean species richness per habitat was highest in the mudflats of the Mira estuary (4.52). In general, higher number of species was found in mudflat and subtidal habitats for all the analysed estuaries. The exception was the Guadiana estuary, where higher mean species richness was found in the saltmarsh habitat (3.10), followed by the subtidal (2.57) and mudflat (2.24) habitats. The variation in species richness in a habitat was also higher in estuaries with higher mean species richness such as the Mira and Sado estuaries (Fig. 2a).

Total mean densities of fish in the different habitats were generally higher in the Mira and Tejo estuaries, with highest mean densities caught in mudflats of the Mira and subtidal habitats of Tejo estuary (52.49 individuals.1000 m^{-2} and 49.01 individuals.1000 m^{-2}, respectively). Variation of these values was also higher in these estuaries. Mean density of fish showed strong variations among the three habitats of the analysed estuaries and no clear pattern of higher values associated with a certain habitat was observed (Fig. 2b).

3.3. Model calibration

In general, GAM built for individual estuaries varied in their success in predicting species richness (Table 2). The models explained 54.5% of the deviance in predicting fish species richness for Guadiana, followed by 48.5%, 42.3%, 38.8% and 32.0% for the Sado, Tejo, Ria Aveiro and Mira estuaries, respectively (Table 2). The combinations of the predictor variables included in the chosen models, for species richness in the estuaries did not differed markedly. Variables such as habitat area and distance to the estuary's mouth significantly explained the variation of species richness in the estuary and were included in most models. Final models for Sado and Mira estuaries included also the percentage of mud in the sediment, while for Ria Aveiro the "season" variable was also included in the final model (Table 2).

The final models built for individual estuaries considering species density as response variable explained higher percentages of deviance: highest for Ria Aveiro (73.0%) and Sado (59.6%) estuaries, followed by Guadiana, Mira and Tejo estuaries (respectively 56.4%, 51.3% and 47.2%) (Table 2). The combination of significant predictor variables in final models, for the several estuaries differed. Nevertheless, predictors such as, "habitat type", distance to estuary's mouth and depth were present in most final models (Table 2).

When using the GAM to predict large scale variation in fish species richness and density values, i.e. using predictor and response datasets pooled from all the analysed estuaries together, the percentages of deviance explained by the final models were lower: 24.5% for species richness and 30.7% for fish density models. The predictors retained in these two models differed, and also differed from those included in the final individual estuary models. The models for species richness in all estuaries also included habitat area and distance to estuary mouth (as did those for individual estuaries) and depth and percentage of organic matter in the sediment (Table 2). The model for species density in all estuaries included some of the estuaries' features (river flow, estuary width, anthropogenic pressure index and habitat complexity) and also some of the variables which were measured locally (distance to estuary mouth, depth, percentage of mud and organic content in the sediment) (Table 2).

3.4. Model validation

The application of each model to the remaining unused 25% of the original dataset yielded generally reasonable results, showing that some of the chosen models were robust when applied to different data (Table 2). Overall, the models were more robust when

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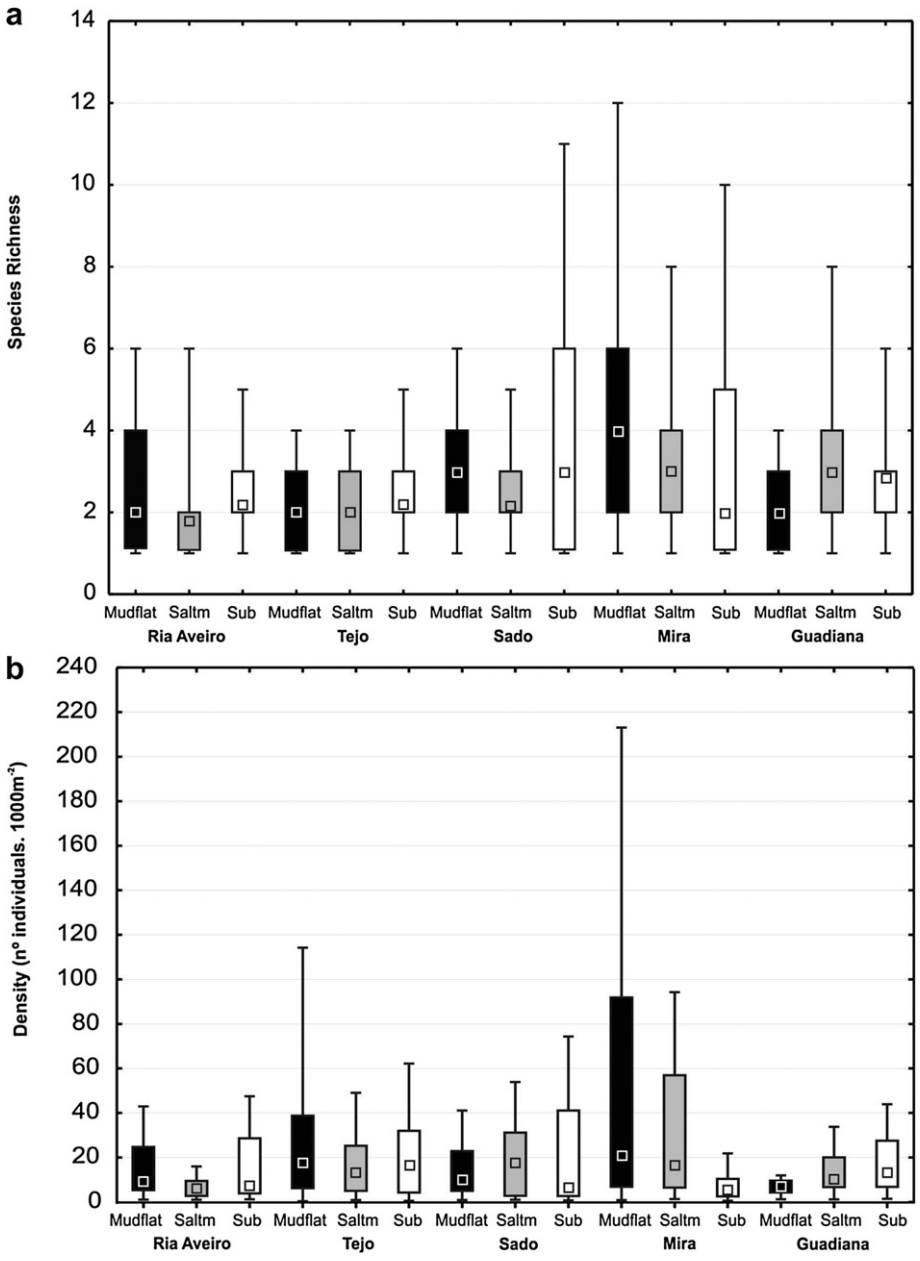


Fig. 2. Variation of fish (a) species richness and (b) densities in the three sampled habitats in estuaries (□ median; ▯ quantiles: 25% and 75%; ⊥ maximum and minimum).

predicting the variation in species densities within each estuary (deviance explained varied from 23.2% to 63.2%, with mean value of 41.7) than when predicting fish species richness variation (deviance explained varied from 17.2% to 62.1%, mean value of 30.5)(Table 2).

Robustness of the model was higher for the Mira estuary when predicting variation of species richness (62.1% of the deviance explained) and for Ria Aveiro when predicting variation of fish densities (63.2% of the variance explained). On the other hand, the chosen model for the Guadiana estuary was the less robust in predicting the variation of species richness (11.2% of deviance explained) and for the Sado estuary for variation of fish species densities (23.2% of deviance explained) (Table 2).

When remaining datasets for fish species richness and density were applied to the large scale models, built with environmental predictors from all the estuaries, the overall robustness of the models was slightly lower.

3.5. Environmental variables

Percentage of mud in the sediment, depth and distance to the estuary mouth, were the significant environmental variables retained in most of the final GAM fitted to individual estuaries, both for fish species richness and density. For instance, the partial effect of the percentage of mud in the sediment in the variation of fish species richness presented a similar pattern for the estuaries in which this variable was important, with higher number of species obtained for higher percentages of mud in the sediment (Fig. 3a). The partial effect of the percentage of mud in the sediment on the variation of fish species densities in the different estuaries varied: for the Tejo estuary, higher densities were registered for sediments comprised by higher percentages of mud, in Sado estuary fish densities decreased until the value of 50% of mud in the sediment, after which increased reaching a maximum around 80%. For the

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Table 2
Final generalized additive models (GAM) built for each estuary and for all estuaries combined, considering as response variables: a) fish species richness; and b) species density. Combination of significant predictor variable, percentage of deviance explained and estimated degrees of freedom (edf) for smoothing parameters of each model are shown both for calibration and validation phases. Abreviations of predictors are: season – season (categorical), area – area, river flow – rflow, estuary mouth width – width, estuary's habitat complexity – comp; habitat type – habitat (categorical), habitat area – habitat area, distance to estuary mouth – dist, depth – depth, organic matter content in the sediment – om, mud content in the sediment – mud.

Response variable	Estuary	Model	Calibration		Validation	
			Dev. Expl (%)	Edf	Dev. Expl (%)	Edf
Species richness	Ria de Aveiro	Season + s(dist)	38.8	1.86	40.4	1.0
	Tejo	Area + s(depth)	42.3	2.7	17.2	2.8
	Sado	Habitat area + s(dist)+s(mud)	48.5	1.1; 1.9	21.4	1.0; 2.0
	Mira	Habitat area + s(dist)+s(mud)	32.0	2.0; 1.5	62.1	1.7; 1.9
	Guadiana	Habitat area + s(dist)	54.5	2.0	11.2	1.1
	All estuaries	Habitat area + s(dist) + s(depth) + s(om)	24.5	1.9; 1.2; 1.9	10.3	1.8; 1.0; 1.0
Density	Ria de Aveiro	Season + habitat + area + s(om)	73.0	1.1	63.2	1.8
	Tejo	s(dist) + s(mud)	47.2	1.6; 1.0	46.6	1.8; 1.6
	Sado	Habitat + dist + s(mud)	59.6	2.5	23.2	2.8
	Mira	s(dist) + s(depth) + s(mud)	51.3	1.1; 2.6; 2.2	52.3	1.9; 1.9; 2.7
	Guadiana	Habitat area + s(dist) + s(om)	56.4	1.8; 2.0	23.4	1.9; 2.0
	All estuaries	rflow + width + press + comp + habitat area + s(dist) + s(depth) + s(om) + s(mud)	30.7	1.9; 1.1; 1.3; 1.8	15.1	2.0; 1.0; 1.7; 1.6

Mira estuary there was a decrease in fish species densities around 80%–90% of mud content in the sediment. When all estuaries were analized together, highest values of fish densities were registered for sediments composed by 50–60% of mud. (Fig. 3b). The scale (range of *Y*-axis) of this partial effect, and hence its importance, was larger for the model of Sado estuary, where the effect of this variable on variation of species richness was stronger.

The partial effect of depth on variation of fish densities also varied: in Mira there was an increase in fish densities between 1 and 2 m of depth, followed by a slight decrease and a new peak at higher depths, while for all the estuaries analysed together, fish abundance decreased from shallow to deeper waters (Fig. 4). The scale of the depth effect was more important in Mira estuary (Fig. 4).

4. Discussion

During the past decade, species predictive models have matured and become a valuable tool for many applications (Corsi et al., 1999; Margules and Pressey, 2000; Miller et al., 2004; Rodriguez et al., 2007). Understanding the factors that cause fluctuations in the population density of organisms has long been a primary goal of ecology. Such fluctuations may be originated by density-dependent mechanisms or density-independent factors, mostly climatic forcing (Maes et al., 2004). In estuarine ecology, density-independent factors have been frequently used to explain spatial and temporal distribution patterns, as estuarine systems are considered a demanding environment for the organisms that they harbour (Day et al., 1989).

GAM have been considered to provide good stability and performance due to the flexible non-parametric or semi-parametric framework which is applied to model the relationship between a response and one or more predictor variables (Hastie and Tibshirani, 1990), making them easier to interpret when compared with other currently available modelling techniques [e.g. generalized linear models (GLM), classification tree analysis and artificial neural network].

In general, dynamics of estuarine systems are governed by non-linear interactions of environmental factors (Ysebaert et al., 2002), thus GAM allow fitting realistic descriptions of relationships between species and environmental predictors with complex and non-linear response curves (Hastie and Tibshirani, 1990; Austin, 2007) as in the present study. GAM have been previously used to predict the distribution of fish species along rivers and estuaries (Kupschus, 2003; Maes et al., 2004; Lassalle et al., 2008; Bacheler et al., 2009), to evaluate estuarine habitat suitability for species that use the estuary as a nursery area (Stoner et al., 2001; Kupschus, 2003) and to predict changes in species distribution patterns with climate change (Lassalle et al., 2009), and is now recognized as a versatile method for species modelling (Guisan and Zimmermann, 2000; Guisan et al., 2002; Segurado and Araujo, 2004). Moreover, skewed response curves are frequently observed as described in ecological theory (Ejrnaes, 2000; Rydgren et al., 2003). It has become clear that there is no standard for current best practice when modelling species environmental/geographical distribution (Austin, 2007) as numerous incompatibilities between ecological data and statistical models can be identified.

GAM were applied to describe the relationship between the variation in fish species richness as well as densities, in several estuarine habitats, and several habitat features and locally measured environmental variables. Overall, the models were successfully built and performed well, explaining a large percentage of deviance in the variation of fish species richness and their densities in estuarine habitats for each analysed estuary. The percentages of deviance explained by the models were equivalent to values obtained in previous studies, for example Lassalle et al. (2009) that used GAM to predict the distribution of diadromous fish species in rivers, or higher than values obtained by Kupschus (2003) when predicting a single fish species abundance in several estuaries.

There is a good knowledge on the processes that govern the relationship between species and the estuarine environment (Whitfield and Harrison, 2003; Akin et al., 2005; Bacheler et al., 2009; Vasconcelos et al., 2010; França et al., 2011) and it can be used to choose potential variables to describe species distribution (Huntley et al., 1995; Guisan and Zimmermann, 2000). The relatively large amount of deviance explained in the present study can be at least partially attributed to an appropriate choice of predictors, as they may represent habitat requirements of species. The combination of variables retained by the models predicting fish species richness and density variation in estuaries presented some similarities.

The distance to estuary mouth was found to be one of the best predictors of the variation of species richness within most of estuaries. This variable acts mostly as a proxy for the salinity, which has been considered an important factor structuring estuarine fish

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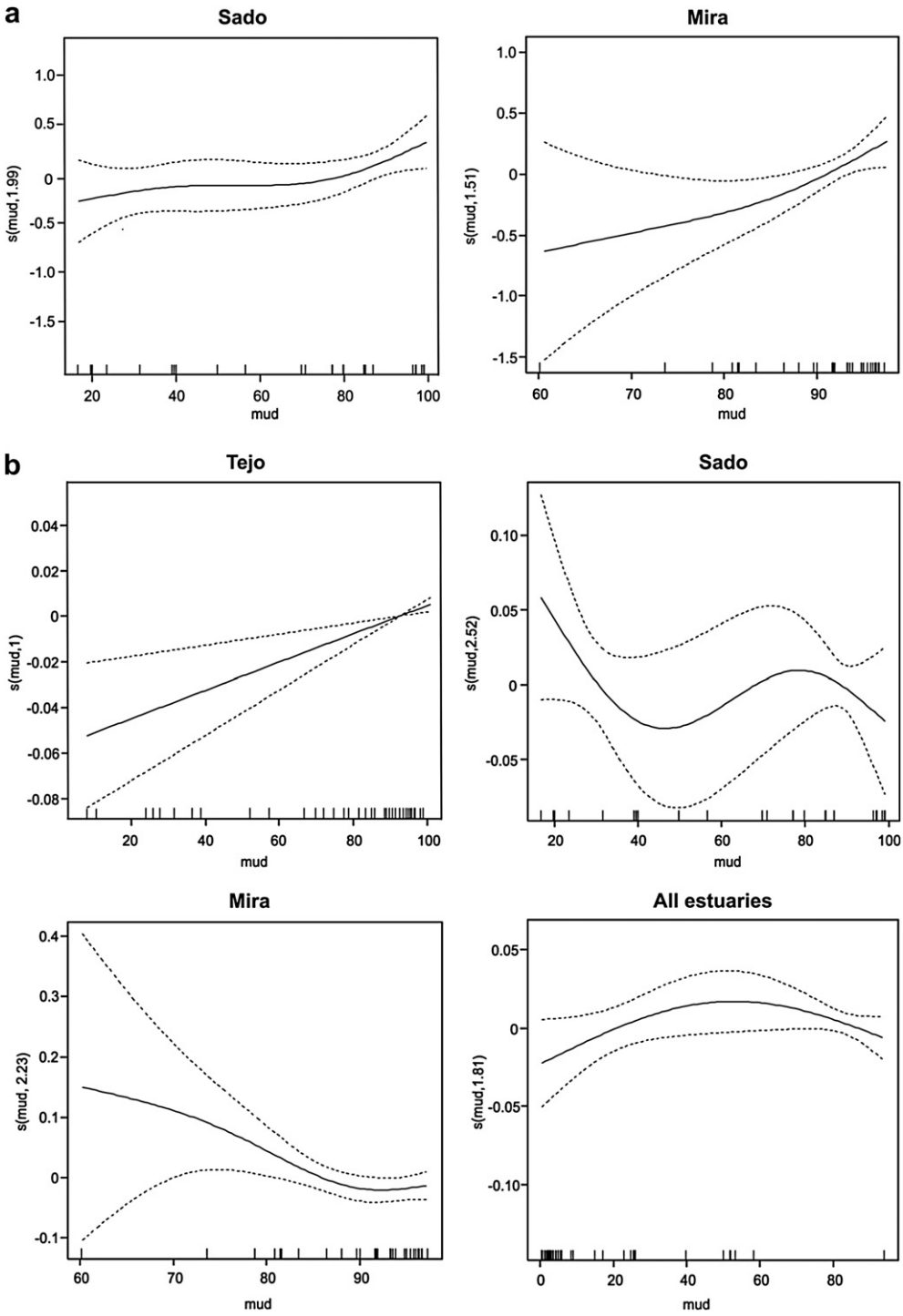


Fig. 3. Partial effect of "percentage of mud in the sediment" on (a) fish species richness and (b) densities, for the estuaries where this predictor was retained in the final GAM. *X*-axes tick marks show the distribution of the predictor values. *Y*-axes represent the partial effect of the predictor in the response variable.

communities (Meng et al., 2001; Lobry et al., 2003; Martino and Able, 2003; Greenwood, 2007; Vasconcelos et al., 2010; França et al., 2011). The occurrence of most estuarine organisms, hence estuarine biodiversity, relates well to spatial and seasonal changes in salinity (Attrill, 2002) and thus represents a density-independent factor which in part explains the presence of estuarine species (Maes et al., 2004). In the present study the seasonality of the sampling did not allow to have temperature and salinity measured continuously, thus these variables were not individually included in the model, but their effect was revealed through variables such as season and distance to estuary mouth. Moreover, the two sampled areas within each estuary were considered replicates as they presented similar environmental variables, thus it was not expected marked variations in salinity which means that probably the importance and significance of this variable is related with the seasonal effect of variation in salinity. The "season" effect was also significant in predicting the variation of fish densities in the estuaries of Ria Aveiro and Sado.

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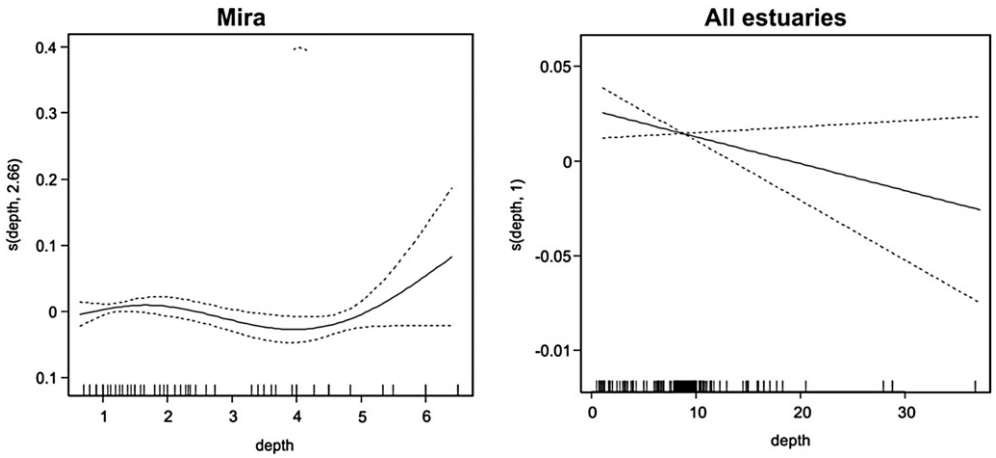


Fig. 4. Partial effect of "depth" on fish species densities for the estuaries where this predictor was retained in the final GAM. *X*-axes tick marks show the distribution of the predictor values. *Y*-axes represent the partial effect of the predictor in the response variable.

In addition to these environmental gradients, estuaries are composed of many different habitat types which have important roles for the whole ecosystem functioning, namely acting as nurseries for different fish species (França et al., 2009). In the present work, sampling design aimed at assessing if habitat was an important feature in the variation of species fish richness and density within each estuary.

Model results showed that the area of habitat was an important and significant variable retained by most of the models explaining variation of fish species richness in most estuaries, while habitat type was important to predict variation in fish densities in estuaries such as Ria Aveiro and Sado. These results highlight the importance of having both habitat complexity and spatial heterogeneity in estuaries, in order to maintain healthy and productive estuarine ecosystems (Irlandi and Crawford, 1997; Martino and Able, 2003; Peterson, 2003). Peterson (2003) emphasized the importance of the habitat mosaic particularly the spatial and seasonal connectivity of different habitat types in the overall estuarine productivity. Even though worldwide numerous estuarine habitats have been degraded to some extent the influence of habitat diversity within estuarine systems has received little attention in the past (Martino and Able, 2003) and only recently has this situation changed. The main goal of these studies is to assess the functional significance of degradation in coastal ecosystems, namely changes in role as nursery or feeding grounds and migratory routes for fish species due to changes in habitat quality and quantity (Peterson, 2003).

In order to evaluate if these models provide accurate hypotheses that sustain reliable predictions of biological patterns (Guisan and Zimmermann, 2000), a data-splitting method was used, with the data randomly split into two: a calibration dataset and an evaluation dataset (Austin, 2007). The model's performance with independent test data, shows its ability to generalize, and indicates the robustness of the model for a given system, rather than its performance using a set of data, which simply indicates its ability to fit to the sample (Tan et al., 2006). The fitted models were considered robust, as the percentage of deviance explained when using the validation dataset was relatively high. This is important as an ideal empirical model should be able to predict not only the outcome of events (samples) on which it was built, but also to predict the states of the system in the face of events which has not be encountered in the finite size samples (Hastie et al., 2001; Tan et al., 2006).

In the present models, environmental variables and estuarine features were included as predictors of fish species richness and density variation. Previous studies have shown that the abundance of fish in estuaries could not be predicted using physico-chemical variables alone, as it is affected by a complex set of events within the estuary (Martino and Able, 2003; Maes et al., 2004), including biotic interactions, such as predation and inter- and intraspecific competition (Wilson, 1991; Olafsson et al., 1994). While the importance of these processes is widely recognized, their relevance in modelling the spatial distribution of species has had only limited examination (Austin, 2007). These biotic variables are often not examined and are frequently excluded from habitat models because of a lack of data (Brown et al., 2000). For models of habitat suitability that use changes in fish density as a proxy for habitat quality, with increasing density indicating habitat suitability, a certain amount of spatial variability is therefore often left unexplained (Eastwood et al., 2001). Despite the strong influence of abiotic processes, the distribution of prey has been identified as a major determinant of estuarine fish habitat use in the situations where it has been examined (Nicolas et al., 2007; Bacheler et al., 2009; Vasconcelos et al., 2010).

Future research is needed before these models and predictions can be applied to evaluate the effects of different management measures within estuaries. This will require an evaluation of the robustness of the models, through the investigation of their applicability to other estuarine systems. The inclusion of new variables and data namely on ecological processes, such as feeding habits, will certainly improve the quality of the models and expand the range of their applicability. Environmental conditions in estuaries vary dynamically in space and time, thus predictions made by models will need to become more robust to changes in the system, and should allow the development of models describing the interaction of processes operating at different scales (Wiens, 1989; Thrush, 1991; Schneider, 1994). These predictive modelling approaches can clearly aid conservation managers by providing information on habitat quantity and quality requirements of fish assemblages, and increasing certainty on the effects of different management options.

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